

Module 5(a) - Power Amplifiers

Power Amplifiers: Amplifier terms, Two load lines, Class A Operation, Class B operation, Class B push pull emitter follower, Class C Operation.

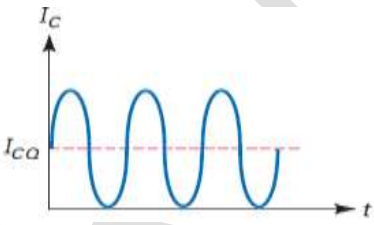
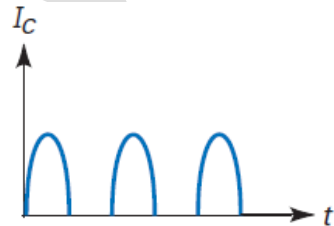
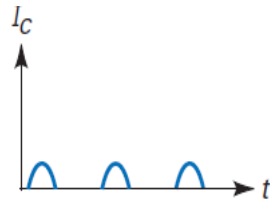
Introduction:

An Amplifier is a device which amplifies the input received from a transducer or from any other input source to a larger version of the input signal to some output device or to another amplifier stage.

Amplifier Terms (Classification of Amplifiers)

There are different ways to classify amplifiers based by their (i) Class of Operation (ii) inter-stage coupling (iii) frequency range (iv) Signal level and so on.

Based on Class of Operation

Parameter	Class A	Class B	Class C
Q point	Located center of the active region and the transistor conducts for full cycle of the input	Located at cut-off so that the transistor conducts for only half cycle of the input	Located below cut-off and transistor conducts for only short duration in half cycle of the input.
Conduction angle	0 to 360° .	0 to 180° .	$0 < \theta < 180^\circ$
Efficiency	25%	78.5%	More than 95%
Collector Current	The input ac base voltage produce collector current which flows for entire input cycle. 	Only the positive half-cycle of ac base voltage produce collector current and is similar to half rectified sine wave. 	Only part of the positive half-cycle of ac base voltage produces collector current in the form of sharp pulses. 
Applications	Used as Preamplifiers in cassette players, CD players	Used in Audio amplifiers, Music players, public addressing systems	Used in radio, TV and Mobile receivers as RF amplifier

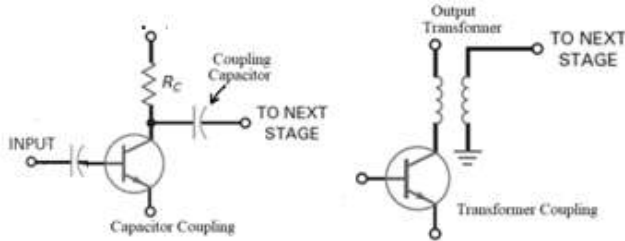
Based on the types of coupling

Coupling is required to connect the input to an amplifier and as well to connect the output of the amplifier to a load. There are two types of coupling namely

(i) AC coupling and (ii) DC coupling

AC coupling - AC coupling is used to block dc voltages and pass ac voltages.

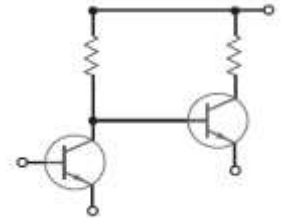
Ex- Capacitive Coupling and Transformer coupling



In **Capacitive coupling** the coupling capacitor connects the ac input to the amplifier as well as transmits the amplified ac voltage to the next stage.

In **Transformer Coupling**, the ac voltage is coupled through a transformer to the next stage.

In **Direct coupling**, there is a direct connection between the collector of the first transistor and the base of the second transistor. Because of this, both the dc and the ac voltages are coupled. Direct coupled amplifiers where there is no lower frequency limit, sometimes called as DC amplifiers. Ex- op-amp.



Based on Frequency of Operation

Based on frequency of operation, basically two types as – Audio frequency (AF) amplifiers and Radio frequency (RF) amplifiers

An **Audio Amplifier** refers to an amplifier that operates in the audio range of 20 Hz to 20 kHz.

A **Radio-Frequency (RF) amplifier** is one that amplifies frequencies above 20 kHz, usually much higher.

Ex- RF amplifiers in AM radios amplify frequencies between 535 and 1605 kHz, and the RF amplifiers in FM radios amplify frequencies between 88 and 108 MHz.

Amplifiers are also classified as **Narrowband** or **Wideband amplifiers**

	Narrowband amplifiers	Wideband amplifiers
1	A narrowband amplifier works over a small frequency range like 450 to 460 kHz.	A wideband amplifier operates over a large frequency range like 0 to 1 MHz.
2	Narrowband amplifiers are usually tuned RF amplifiers , which means that their ac load is a high-Q resonant tank tuned to a radio station or television channel	Wideband amplifiers are usually untuned; that is, their ac load is resistive.

Based on Signal level

In **Small-signal amplifiers**, the peak-to-peak swing in collector current is less than 10 percent of quiescent collector current. Ex – A low noise preamplifier in a stereo system

In **Large-Signal Amplifiers**, peak-to-peak signal uses all or most of the load line. Ex- Power amplifier at the output stage driving a loud speaker.

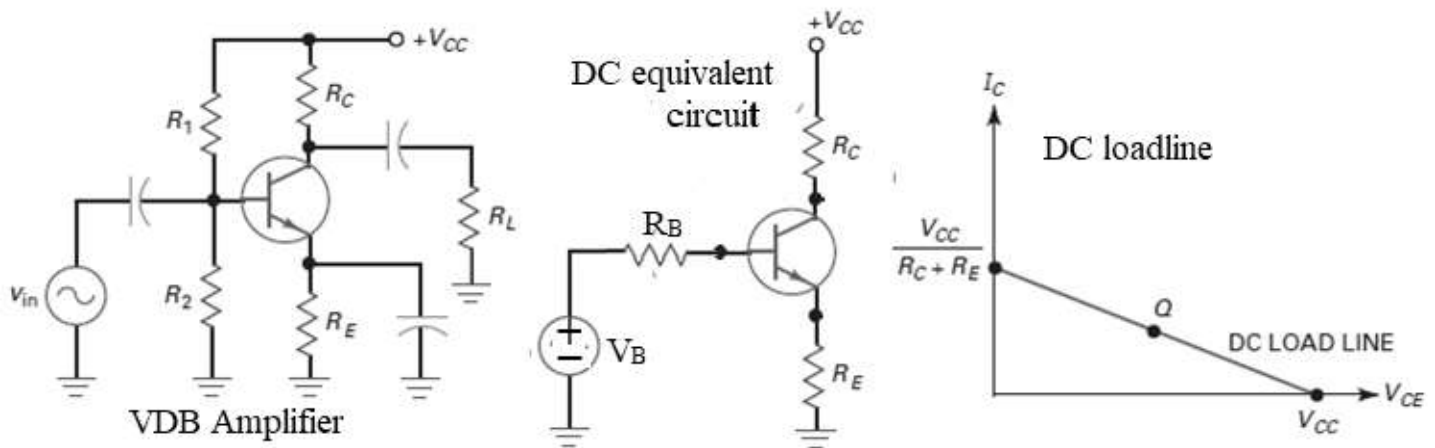
Concept of Two Load Lines

The load line is a straight line drawn on the output characteristics of a transistor used to locate the optimum operating point for the given biasing conditions..

Every amplifier has a dc-equivalent circuit and an ac-equivalent circuit. Because of this, it has two load lines: a dc load line and an ac load line. For large-signal amplifiers, the Q point has to be at the middle of the ac load line to get the maximum possible output swing.

DC load line

Consider a VDB amplifier shown in figure. The base current is controlled by the biasing resistor R_2 . For very large values of R_2 , the transistor goes into saturation while smaller values of R_2 will drive the transistor to cut-off.



Where $R_B = R_1 || R_2$ and $V_B = \frac{V_{CC} R_2}{R_1 + R_2}$

Applying KVL to collector emitter loop

$$V_{CC} - I_C(R_C + R_E) - V_{CE} = 0 \quad \text{or} \quad I_C = \frac{-V_{CE}}{R_C + R_E} + \frac{V_{CC}}{R_C + R_E} \quad \text{----- (A)}$$

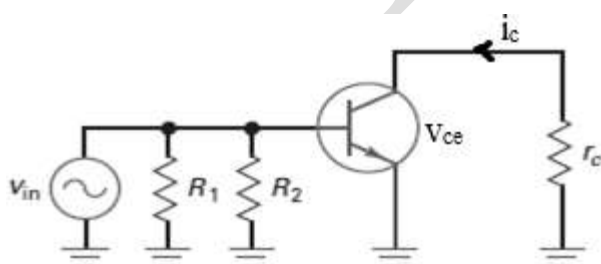
Equation (A) represents a straight line of the form $y = mx + C$ and the end points of the DC load line are determined by

$$I_{C(sat)} = \left[\frac{V_{CC}}{R_C + R_E} \right]_{V_{CE}=0} \quad \text{and} \quad V_{CE(cut-off)} = [V_{CC}]_{I_C=0}$$

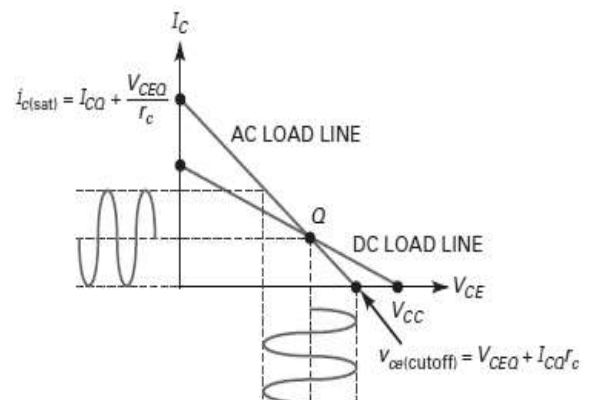
With the emitter at ac ground, R_E has no effect on the ac operation. Also the ac collector resistance is less than the dc collector resistance. Therefore, when an ac signal is applied, the instantaneous operating point moves along the **ac loadline**.

The peak-to-peak sinusoidal current and voltage are determined by the ac load line. It's important to note that the ac and dc load lines intersect at the Q point. For maximum unclipped peak-to-peak ac output voltage, the Q point has to be placed at the center of the ac load line.

The equation for determining end points of the ac load line are obtained by using ac equivalent circuit shown in figure.



KVL to



collector loop yields $v_{ce} + i_c r_c = 0$ and $i_c = -\frac{v_{ce}}{r_c}$ ----- (A) where $r_c = R_C || R_L$

Also $i_c = \Delta I_C = I_C - I_{CQ}$ and $v_{ce} = \Delta V_{CE} = V_{CE} - V_{CEQ}$

Using these values in equation (A) results $I_C - I_{CQ} = -\frac{V_{CE}}{r_c} + \frac{V_{CEQ}}{r_c}$

Rearranging the terms results in the equation of ac load line

$$I_C = I_{CQ} + \frac{V_{CEQ}}{r_c} - \frac{V_{CE}}{r_c} \text{ ----- (B)}$$

When the transistor goes into saturation (at $V_{CE} = 0$), $i_{c(sat)} = I_{CQ} + \frac{V_{CEQ}}{r_c}$ ----- (C)

When the transistor goes into cut-off (at $I_C = 0$), $V_{ce(cut-off)} = V_{CEQ} + I_{CQ} r_c$ ----- (D)

Because the ac load line has a higher slope than the dc load line, the Maximum Peak-to-Peak (MPP) output is always less than the supply voltage $MPP < V_{CC}$

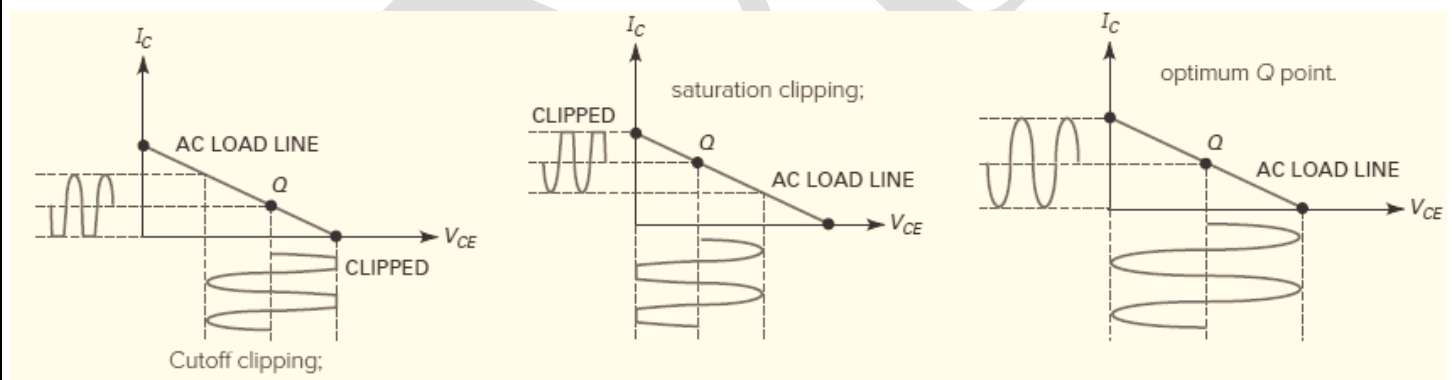
For any Q point, the Maximum Peak o/p is:
MP = $I_{CQ} r_c$ or V_{CEQ} whichever is smaller

The maximum peak-to-peak (MPP) output is: **MPP = 2MP**

The circuit's emitter resistance can be adjusted for the optimum Q point $R_E = \left[\frac{R_C + r_c}{\left(\frac{V_{CC}}{V_E} \right) - 1} \right]$

When the Q point is below the center of the ac load line, $MP = I_{CQ} r_c$.
If the Q point is above the center of the ac load line, $MP = V_{CEQ}$
When the Q point is at the center of the ac load line: $I_{CQ} r_c = V_{CEQ}$

Figure shows the location of Q point and corresponding clipping levels



Problem - Determine the ac load line saturation and cutoff points for the circuit shown below. Also, find the maximum peak-to-peak output voltage.

Solution

$$V_B = \left[\frac{V_{CC} R_2}{R_1 + R_2} \right] = \left[\frac{30(68)}{490 + 68} \right] = 3.7V$$

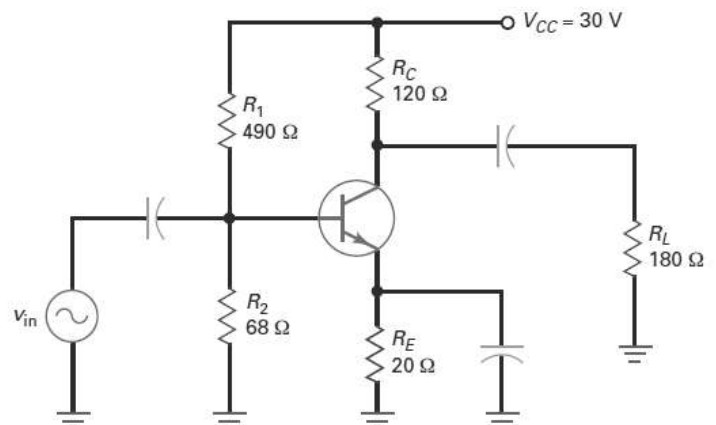
$$V_E = V_B - V_{BE} = 3.7 - 0.7 = 3V$$

$$I_E = \left[\frac{V_E}{R_E} \right] = \left[\frac{3}{20} \right] = 150 \text{ mA} \cong I_{CQ}$$

$$V_{CEQ} = V_{CC} - I_C R_C - V_E = 30 - 0.15(120) - 3V = 9V$$

$$r_c = R_C || R_L = 120 || 180 = 72\Omega$$

$$i_{c(sat)} = I_{CQ} + \frac{V_{CEQ}}{r_c} = 0.15 + \frac{9}{72} = 275 \text{ mA}$$



$$V_{ce(\text{cut-off})} = V_{CEQ} + I_{CQ} r_c = 9 + 0.15(72) = 19.8 \text{ V}$$

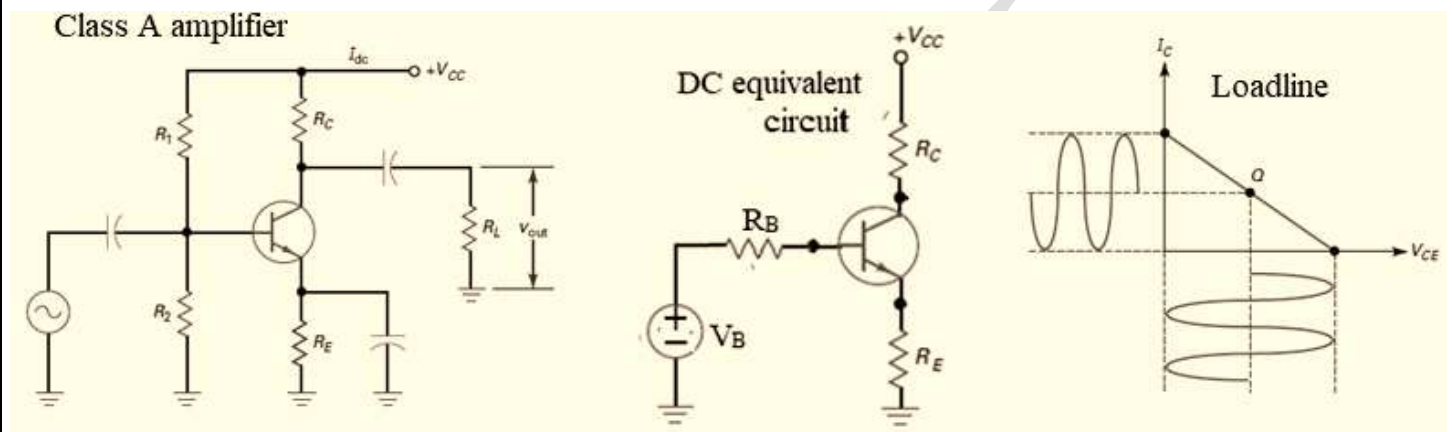
$$I_{CQ} r_c = 0.15(72) = 10.8 \text{ V} > V_{CEQ}$$

Hence MP = $V_{CEQ} = 9 \text{ V}$ and MPP = $2\text{MP} = 18 \text{ V}$

Class-A Amplifier

In Class A power amplifier, the transistor conducts for entire cycle of the input, the operating point Q is located at the center of the active region and the conduction angle is 0 to 360° .

Figure shows the circuit of class A power amplifier using voltage divider bias configuration. DC equivalent circuit and its load line.



DC Analysis:

Under dc conditions, the capacitor acts as open circuit. The DC bias is set by the voltage divider network R_1 - R_2 connected to V_{CC} . Figure shows the dc equivalent circuit, where $R_B = R_1 || R_2$ and $V_B = \frac{V_{CC} R_2}{R_1 + R_2}$

Applying KVL to collector emitter loop

$$V_{CC} - I_C(R_C + R_E) - V_{CE} = 0 \quad \text{or} \quad I_C = \frac{-V_{CE}}{R_C + R_E} + \frac{V_{CC}}{R_C + R_E} \quad \text{--- (A)}$$

Equation (A) represents a straight line of the form $y = mx + C$ and the end points of the DC load line are determined by

$$I_{C(\text{sat})} = \left[\frac{V_{CC}}{R_C + R_E} \right]_{V_{CE}=0} \quad \text{and} \quad V_{CE(\text{cut-off})} = [V_{CC}]_{I_C=0}$$

The dc load line is drawn with end points $(V_{CC}, 0)$ and $\left(0, \frac{V_{CC}}{R_C + R_E}\right)$ with slope $\left(\frac{-1}{R_C + R_E}\right)$ as shown in figure.

Fig shows the DC load line with operating point Q set at optimum level. If the dc bias collector current is set at one half the possible signal swing $\left(0 \text{ and } \frac{V_{CC}}{R_C + R_E}\right)$, the largest collector swing is possible. Similarly if V_{CEQ} is set at one half the supply voltage, the largest voltage swing is possible.

AC operation:

When the input signal applied to the amplifier, the output will vary from its bias levels. A small input signal will cause the base current to vary above and below the Q point; which will then cause the collector current to vary from the set DC bias point (similarly for V_{CE} also). The limits for the current are at 0 amp and V_{CC}/R_C and the limits of V_{CE} are 0 volts and V_{CC} .

Class A Power amplifier Calculations:

Transistor Power Dissipation

With no input signal applied, the quiescent power dissipation $P_{DQ} = V_{CEQ} I_{CQ}$

$$\text{Output power } P_{out} = P_{o(ac)} = \frac{[V_{o(rms)}]^2}{R_L} = \frac{1}{R_L} \left(\frac{V_{o(peak)}}{\sqrt{2}} \right)^2 = \frac{1}{R_L} \left(\frac{V_{o(p-p)}}{2\sqrt{2}} \right)^2 = \frac{[V_{o(p-p)}]^2}{8R_L}$$

$$\text{Similarly AC input power } P_{in} = \frac{[V_{in}]^2}{8Z_{in(stage)}}$$

$$Z_{in(stage)} = R_B \parallel Z_{in(base)} \quad \text{where } R_B = R_1 \parallel R_2;$$

Power Gain - Power Gain is ratio of the ac output power to the ac input power and is given by

$$A_P = \frac{P_{out}}{P_{in}} = \left(\frac{V_{o(p-p)}}{V_{in}} \right)^2 \left[\frac{Z_{in(stage)}}{R_L} \right]$$

The DC Power supplied to an amplifier by the dc source is given by

$$P_{i(dc)} = V_{CC} I_{dc} \quad \text{and } I_{dc} = I_{bias} + I_{CQ} \quad \text{where } I_{bias} = \left[\frac{V_{CC}}{R_1 + R_2} \right]$$

Efficiency

$$\text{Efficiency } n\% = \frac{\text{AC Outpower } P_{out}}{\text{DC input power } P_{i(dc)}} = \frac{[V_{out}]^2}{8R_L} \left(\frac{1}{V_{CC} I_{dc}} \right)$$

Current Drain

The dc voltage source has to supply a dc current I_{dc} to the amplifier. The dc current I_{CQ} is called the **Current drain** of the stage. $I_{drain} = I_{CQ}$

Problem – For the Class A amplifier shown in figure, if the output voltage is 18V(p-p) and input impedance of the base is 100Ω , determine (i) the power gain (ii) Power dissipation and (iii) efficiency

Solution

Given $V_{out} = 18V$; $Z_{in(base)} = 100\Omega$, and $R_B = R_1 \parallel R_2$

$$Z_{in(stage)} = Z_{in(base)} \parallel R_B = 100 \parallel 490 \parallel 68 = 37.4 \Omega$$

(i) Power gain A_P

$$\text{AC output power } P_{out} = \frac{[V_{out}]^2}{8R_L} = \frac{[18]^2}{8(180)} = 225\text{mW}$$

$$\text{AC input power } P_{in} = \frac{[V_{in}]^2}{8Z_{in(stage)}} = \frac{[0.2]^2}{8(37.4)} = 133.7\mu\text{W}$$

$$\text{Power Gain } A_P = \frac{P_{out}}{P_{in}} = \left[\frac{225\text{mW}}{133.7\mu\text{W}} \right] = 1680$$

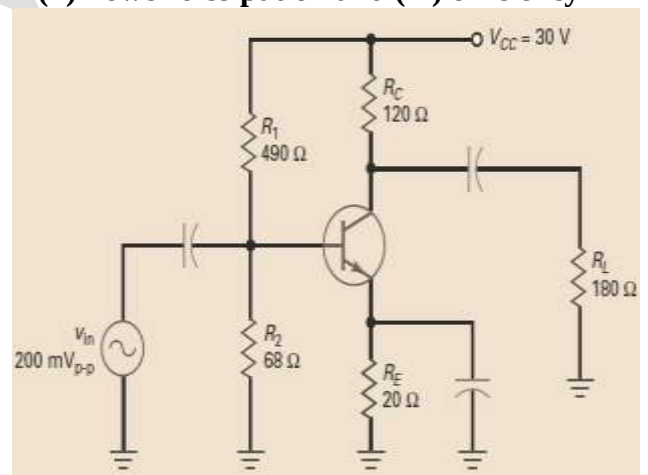
(ii) Power dissipation P_{DQ}

$$V_B = \left[\frac{V_{CC} R_2}{R_1 + R_2} \right] = \left[\frac{30(68)}{490 + 68} \right] = 3.7V$$

$$V_E = V_B - V_{BE} = 3.7 - 0.7 = 3V$$

$$I_E = \left[\frac{V_E}{R_E} \right] = \left[\frac{3}{20} \right] = 150 \text{ mA} \cong I_{CQ}$$

$$V_{CEQ} = V_{CC} - I_C R_C - V_E = 30 - 0.15(120) - 3V = 9V$$



Transistor power dissipation $P_{DQ} V_{CEQ} I_{CQ} = (0.15)(9) = 1.35 \text{ W}$

The minimum bias current $I_{bias} = \left[\frac{V_{CC}}{R_1 + R_2} \right] = \left[\frac{30}{490 + 68} \right] = 53.8 \text{ mA}$

and $I_{dc} = I_{bias} + I_{CQ} = [53.8 + 150] \text{ mA} = 203.8 \text{ mA}$

DC power input $P_{i(dc)} = V_{CC} I_{dc} = (30V)(203.8\text{mA}) = 6.11\text{W}$

Efficiency $n\% = \frac{\text{AC Outpower } P_{out}}{\text{DC input power } P_{i(dc)}} = \left(\frac{225\text{mW}}{6.11\text{W}} \right) \times 100 = 3.18\%$

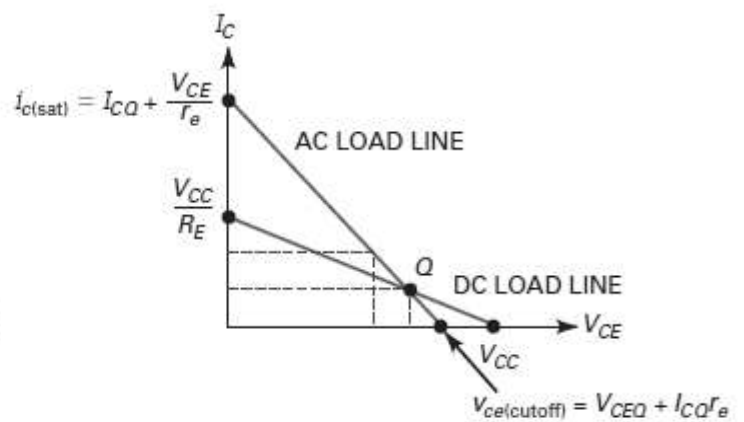
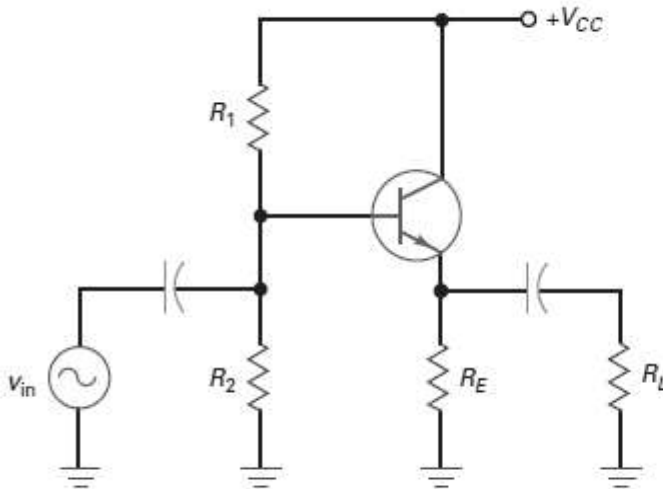
Emitter-Follower Class A Power Amplifier

Figure shows the circuit diagram of class A amplifier using an emitter follower. The Q point is located at the center of the ac load line to get maximum peak-to-peak (MPP) output.

The DC load line end points are given by $I_{C(sat)} = \left[\frac{V_{CC}}{R_E} \right]$ and $V_{CE(cut-off)} = V_{CC}$

The ac emitter resistance is less than the dc emitter resistance. When an ac signal comes in, the instantaneous operating point moves along the ac load line.

The AC load line end points are given by $i_{c(sat)} = I_{CQ} + \left[\frac{V_{CE}}{r_e} \right]$ and $V_{CE(cut-off)} = V_{CEQ} + I_{CQ} r_e$



Problem – For an emitter follower based Class A amplifier has $R_1 = 50\Omega$, $R_2 = 100\Omega$, $R_E = R_L = 16\Omega$ and $V_{CC} = 12\text{V}$. What are the values of I_{CQ} , V_{CEQ} , and r_e . Determine the ac saturation and cutoff points Also, find the circuit's MPP output voltage.

Solution $V_B = \left[\frac{V_{CC} R_2}{R_1 + R_2} \right] = \left[\frac{12(100)}{50 + 100} \right] = 8\text{V}$ $V_E = V_B - V_{BE} = 8 - 0.7 = 7.3\text{V}$

$I_E = \left[\frac{V_E}{R_E} \right] = \left[\frac{7.3}{16} \right] = 456 \text{ mA} \cong I_{CQ}$ $V_{CEQ} = V_{CC} - V_E = 12 - 7.3 = 4.7\text{V}$

$r_e = R_E || R_L = 16 || 16 = 8\Omega$

The AC load line end points are given by $i_{c(sat)} = I_{CQ} + \left[\frac{V_{CE}}{r_e} \right] = 0.456 + \left[\frac{4.7}{8} \right] = 1.04\text{A}$

$V_{CE(cut-off)} = V_{CEQ} + I_{CQ} r_e = 4.7 + 0.456(8) = 8.35\text{V}$

$$I_{CQ} r_e = 0.456(8) = 3.65V < V_{CE}$$

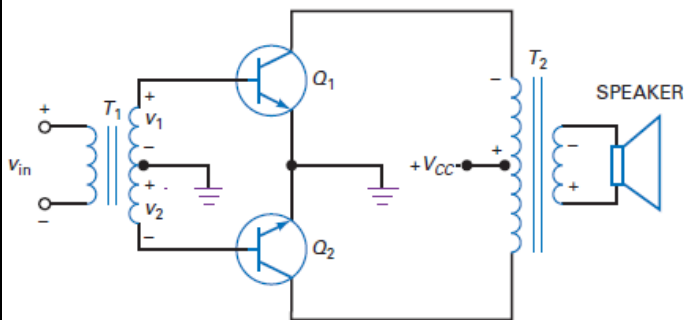
$$\text{hence } MP = I_{CQ} r_e = 3.65V \text{ and } MPP = 2MP = 7.3V$$

Class-B Operation

Class B amplifiers are used in some applications like battery-powered systems where the current drain and stage efficiency become important considerations.

Class B Push-Pull Amplifier

Figure shows a basic Class-B Push pull amplifier which uses two transistors such that one transistor conducts for half a cycle while the other is off, and vice versa, hence the name Push-pull.



Operation

On the positive half-cycle of input voltage, the secondary winding of T_1 has voltage v_1 and v_2 , as shown. Therefore, the upper transistor conducts and the lower one cuts off. The collector current through Q_1 flows through the upper half of the output primary winding. This produces an amplified and inverted voltage, which is transformer-coupled to the loudspeaker. On the next half-cycle of input voltage, the polarities reverse. Now, the lower transistor turns on and the upper transistor turns off. The lower transistor amplifies the signal, and the alternate half-cycle appears across the loudspeaker. Since each transistor amplifies one-half of the input cycle, the loudspeaker receives a complete cycle of the amplified signal.

Advantages and Since there is no bias, each transistor is at cutoff when there is no input signal, an advantage because there is no current drain when the signal is zero.

Another advantage is improved efficiency where there is an input signal. The maximum efficiency of a Class-B push-pull amplifier is 78.5 percent.

Disadvantages

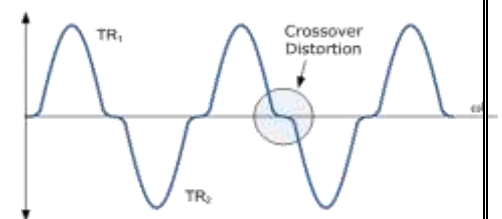
The main disadvantage of the amplifier is the use of transformers. Audio transformers are bulky and expensive. There exists cross over distortion in the output waveform

Cross Over Distortion

For transistors to conduct, the EB junction has to be forward biased. The applied input voltage must be greater than the V_{BE} of the transistor (0.7V for Si) Hence the output current is zero for the period $0 < V_{in} < V_{BE}$.

This results in discontinuity in the output waveform and this distortion is known as cross over distortion.

Cross over distortion can be minimized by biasing the transistors slightly above cut-off and the scheme is known as Class AB amplifier

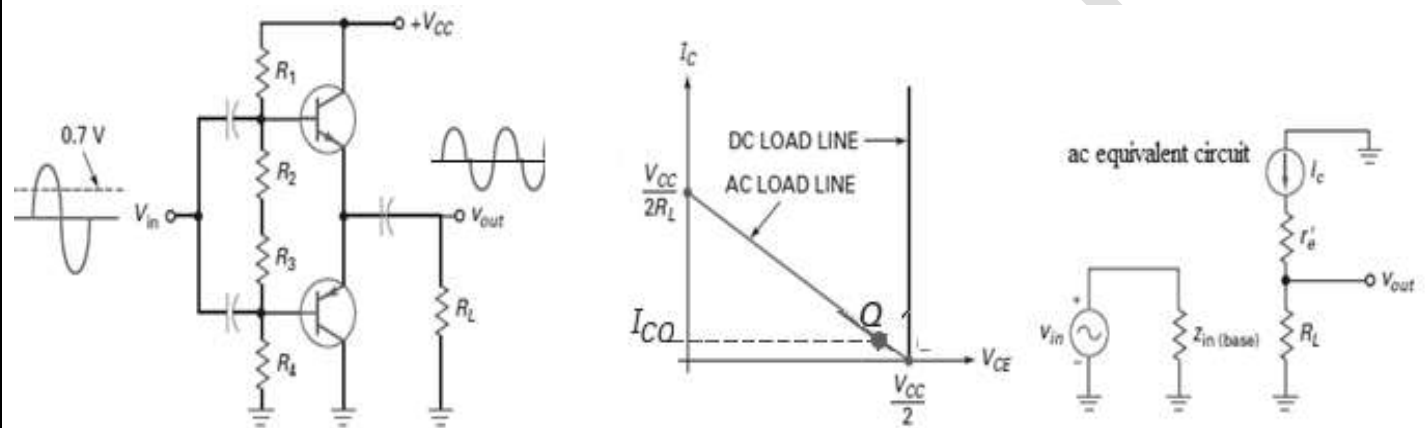


Class-B push-pull emitter follower (Class AB amplifier)

Figure shows a Class-B push-pull amplifier using npn and pnp emitter follower connected in a push-pull arrangement. The biasing resistors are selected to set the Q point at the verge of cutoff by biasing emitter diode of each transistor between 0.6 and 0.7 V, so that it is on the verge of conduction and $I_{CQ} = 0$ and $V_{CEQ} = (V_{CC}/2)$

Operation :- On the positive half-cycle of input voltage, the upper transistor conducts and the lower one cuts off. The upper transistor acts like an ordinary emitter follower, so that the output voltage approximately equals the input voltage. On the negative half-cycle of input voltage, the upper transistor cuts off and the lower transistor conducts. The lower transistor acts like an ordinary emitter follower and produces a load voltage approximately equal to the input voltage. The upper transistor handles the positive half-cycle of input voltage, and the lower transistor takes care of the negative half-cycle.

Since there is no dc resistance in the collector or emitter circuits, the dc load line is vertical, With the



application of input, when either transistor is conducting, its operating point moves up along the ac load line. The voltage swing of the conducting transistor can go all the way from cutoff to saturation. Hence the maximum peak-to-peak output is: $MPP = V_{CC}$

The voltage gain of the emitter follower is $A_v \cong 1$ and the input impedance of the base is: $Z_{i(base)} \cong \beta R_L$

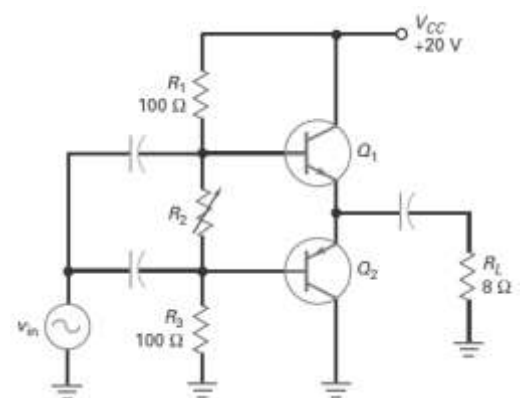
AC output power $P_{out(ac)} = \frac{[V_{out}]^2}{8R_L}$	Maximum o/p power $P_{out(max)} = \frac{[MPP]^2}{8R_L}$	Maximum transistor power dissipation $P_{D(max)} = \frac{[MPP]^2}{40R_L}$
AC input power $P_{in} = \frac{[V_{in}]^2}{8Z_{in(stage)}}$		
$P_{I(dc)} = V_{CC} I_{dc}$	Efficiency $\eta\% = \frac{P_{out(ac)}}{P_{I(dc)}} \times 100$	Power Gain $A_p = \frac{P_{out}}{P_{in}}$

Problem:- For the amplifier circuit shown, (a) calculate the maximum transistor power dissipation and maximum output power. (b) If the adjustable resistance $R_G = 15\Omega$, calculate the efficiency of the amplifier.

Solution

$MPP = V_{CC} = 20V$; $R_L = 8\Omega$; $V_{CEQ} = (V_{CC}/2) = 10V$

Maximum transistor power dissipation $P_{D(max)} = \frac{[MPP]^2}{40R_L}$



$$P_{D(max)} = \frac{[20]^2}{40(8)} = 1.25 \text{ W}$$

$$\text{Maximum output power } P_{out(max)} = \frac{[MPP]^2}{8R_L} = \frac{[20]^2}{8(8)} = 6.25 \text{ W}$$

$$I_{bias} = \frac{V_{CC}}{R_1 + R_G + R_2} = \frac{20}{100 + 15 + 100} = 0.093 \text{ A} \quad i_{c(sat)} = \frac{V_{CEQ}}{R_L} = \frac{10}{8} = 1.25 \text{ A}$$

$$\text{Average value of collector current } I_{av} = \frac{i_{c(sat)}}{\pi} = 0.398 \text{ A}$$

$$I_{dc} = I_{av} + I_{bias} = 0.093 + 0.398 = 0.491 \text{ A} \quad \text{and} \quad \text{input power } P_{i(dc)} = V_{CC} I_{dc} = 20(0.491) = 9.82 \text{ W}$$

$$\text{Efficiency } \eta\% = \frac{P_{out(ac)}}{P_{i(dc)}} \times 100 = \frac{6.25}{9.82} \times 100 = 63.6\%$$

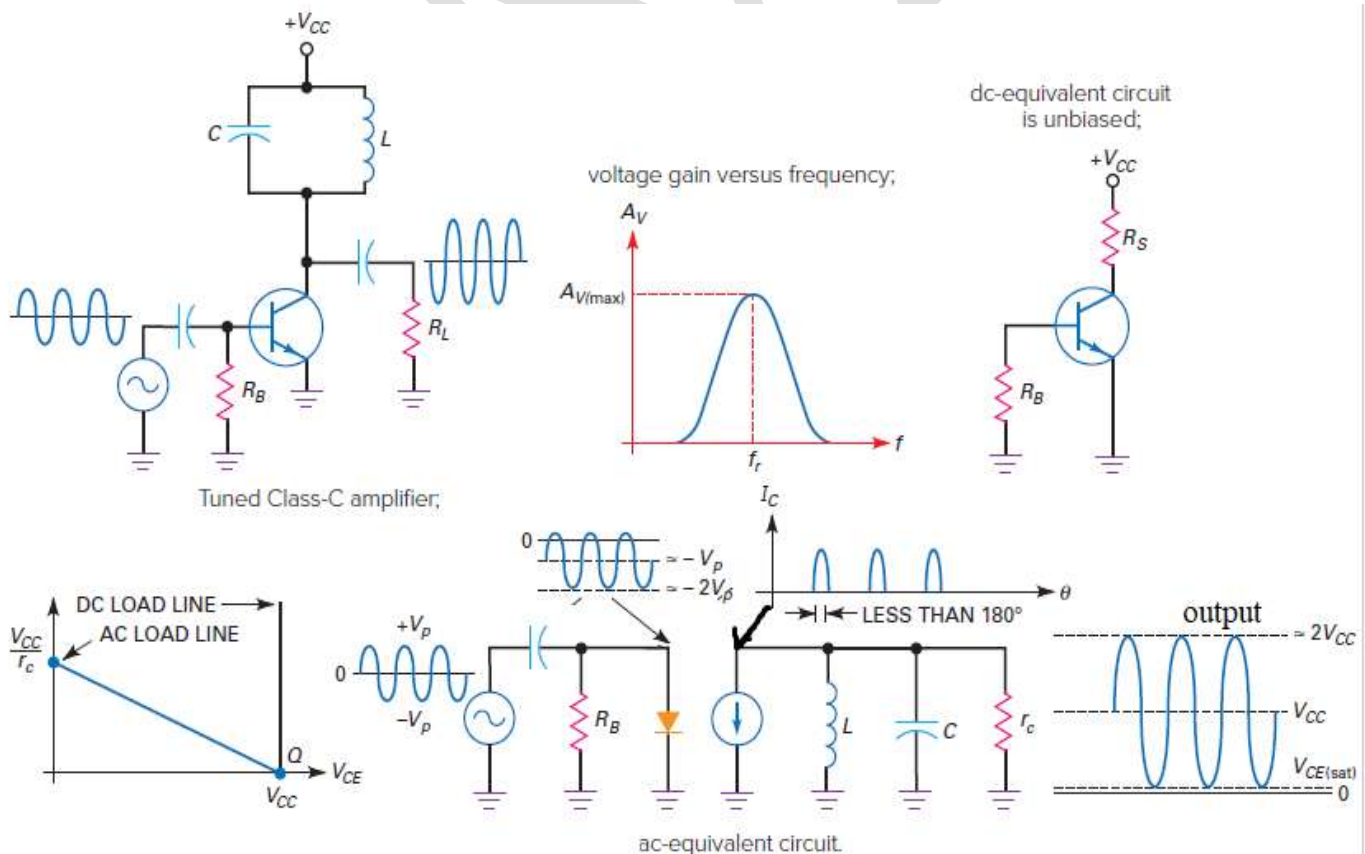
Class-C Amplifier

Introduction - Class-C amplifiers use resonant circuits for the load and are called as tuned amplifiers. With Class-C operation, the Q point is located below cut-off and the collector current flows for less than half a cycle. Hence the conduction angle is less than 180° .

Working

Figure shows the circuit diagram of Class C tuned amplifier with ac, dc equivalent circuits and frequency response. The ac input voltage drives the base, and an amplified output voltage appears at the collector. The amplified and inverted signal is then capacitively coupled to the load resistance. Because of the parallel resonant circuit, the output voltage is maximum at the resonant frequency.

$$f_r = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$



The Class-C amplifier is unbiased and the resistance R_S in the collector circuit is the series resistance of the inductor which very small, hence the DC load line is vertical.

The Q point is at the lower end of the ac load line. When an ac signal is present, the instantaneous operating point moves up the ac load line toward the saturation point. The maximum pulse of collector current is given by the saturation current (V_{CC}/r_c) **where $r_c = R_S || R_L$**

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DC Clamping of Input Signal

Consider the ac-equivalent circuit. The input signal drives the emitter diode, and the amplified current pulses drive the resonant tank circuit. The input capacitor is part of a negative dc clamper, the signal appearing across the emitter diode is negatively clamped. Only the positive peaks of the input signal can turn on the emitter diode. For this reason, the collector current flows in brief pulses (non sinusoidal waveform) which is rich in Harmonics.

Filtering the Harmonics

The parallel resonant tank circuit has a high impedance only at the fundamental frequency. This produces a large voltage gain at the fundamental frequency and very little voltage gain at other harmonic frequencies. Hence acts like filter by removing the harmonics to produce a sine wave at the output.

Advantages of Class C power amplifier.

(i) High efficiency. (ii) Excellent in RF applications. (iii) Lowest physical size for a given power output.

Disadvantages of Class C power amplifier.

(i) Lowest linearity. (ii) Not suitable in audio applications. (iii) Creates a lot of RF interference. (iv) It is difficult to obtain ideal inductors and coupling transformers. (v) Reduced dynamic range.

Applications of Class C power amplifier. (i) RF oscillators. (ii) RF amplifier. (iii) FM transmitters. (iv) Booster amplifiers. (v) High frequency repeaters. (vi) Tuned amplifiers etc.

Review Questions

1. What is an amplifier? Explain the classification of amplifiers based on Class of Operation.
2. What is an amplifier? Explain the classification of power amplifiers.
3. Write short notes on Types of coupling used in amplifiers.
4. What is a load line? Draw and explain the concept of DC load line and AC load line for a transistor using Voltage Divider Bias (VDB) showing the end points.
- 5 What is a Class A power amplifier? Explain the operation of Class A amplifier and derive the expression for its power gain, power dissipation, current drain and its efficiency.
6. With a neat circuit diagram, explain the operation of Class B push-pull amplifier.
7. What is cross over distortion? Explain how it can be eliminated in Class B push-pull emitter follower configuration?
8. What is a Class C Power amplifier? With a neat circuit diagram explain its operation and also the DC clamping and harmonics filtering in Class C operation.